INDIAN INSTITUTE OF TECHNOLOGY BOMBAY

Department of Mathematics

SI 427 (Probability I)

Tutorial Sheet-IX

1. Let X and Y be discrete random variables such that $P\{|X - Y| \leq M\} = 1$ for some $M > 0$. Show that if Y has finite mean, then X has finite mean.

2. Give an example of a discrete random variable with finite mean but with infinite variance.

3. Let X be a discrete random variable with finite second moment. Show that it has finite mean.

4. Let X be standard normal random variable. Find EX^2.

5. Let X be exponentially distributed with parameter λ. Find the 4th moment.

6. Let $\varphi : \mathbb{R} \to \mathbb{R}$ be a continuous bounded function and X be a random variable with pdf f. Show that

$$E[\varphi(X)I_{\{X\leq 0\}}] \;=\; \int_{-\infty}^{0} \varphi(x)f(x)dx.$$

7. Let X and Y be independent and identically distributed exponential random variables with parameter λ. Find the mean of $\max\{X, Y\}$.

8. Let X, Y be random variables with joint pdf f. Without using the theorem on computing $E[\varphi(X, Y)]$ where φ is a continuous function, show that

$$E[XY] \;=\; \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xyf(x, y)dxdy.$$

9. Let X be Binomial (p). Find $\{a_n\}$ where $a_n = EX^n, n \geq 1$.

10. Let X be Binomial (n, p) with $EX = 1$ and $EX^2 \;=\; 2$, find the pmf of X.

11. Let X, Y has joint pdf

$$f(x,y) = \begin{cases} 2 & \text{if} \quad 0 \le x \le y \le 1 \\ 0 & \text{if} \quad \text{otherwise.} \end{cases}$$

    Find the covariance matrix of (X, Y).

12. Find all median/(s) of Cauchy random variable.